

Fin-Cash



Capstone SRS Template 2025-2026

Team Code: 13

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Software Requirements Specification (SRS)

1. Introduction

1.1 Purpose

The system aims to provide a centralized platform for managing, analyzing, and visualizing financial data. It reduces the need for manual spreadsheet processing, automates financial report generation, and provides interactive dashboards and forecasting tools to support efficient and informed financial decision-making.

1.2 Scope

• Key Features:

- Secure user authentication and Role-Based Access Control (RBAC).
- Importing and validating financial data from CSV and XLSX files.
- Processing and aggregating financial data using predefined calculation rules.
- Interactive dashboards for monitoring key financial KPIs.
- Automated generation of financial reports, including income statements, balance sheets, and cash flow reports.
- Financial forecasting based on historical financial data.
- Exporting reports and analytics in PDF, XLSX, and CSV formats.

Activity logging for security and auditing purposes.

- **Target Users:** Finance staff, financial analysts, administrators, and managers/executives.
- **Limitations:** The system focuses on financial analytics, reporting, visualization, and forecasting. It does not perform direct banking transactions or replace existing accounting systems. Integration with external legacy databases may be considered as a future enhancement.

1.3 Definitions and Acronyms

KPI (Key Performance Indicator): A measurable value used to monitor and evaluate financial performance.

RBAC (Role-Based Access Control): A security mechanism that manages user access and permissions based on assigned roles.

Dashboard: A visual interface that displays key financial metrics, charts, and performance indicators.

Forecasting: The process of estimating future financial trends or values using historical data.

Financial Report: A structured document that presents financial information, such as revenue, expenses, profit, and cash flow.

Data Validation: The process of checking imported financial data for missing, invalid, or incorrectly formatted values.

CSV (Comma-Separated Values): A file format used to store structured data in a tabular format.

XLSX: A Microsoft Excel file format used to store structured spreadsheet data.

Audit Log: A record of important user activities and system events used for monitoring and security auditing.

2. Overall Description

2.1 Problem Statement

Financial teams often rely on scattered spreadsheets and manually processed financial data, which can lead to calculation errors, duplicated work, and delays in preparing financial reports. Managers and executives may also lack a centralized view of important financial KPIs, making it difficult to monitor financial performance efficiently.

Fin-Cash addresses these challenges by providing a centralized web-based platform for importing, processing, analyzing, and visualizing financial data. The system automates financial report generation, provides interactive dashboards, supports financial forecasting, and applies Role-Based Access Control (RBAC) to ensure that users can access only the features and data appropriate to their roles.

2.2 Design Thinking Process Summary

- **Empathize:** Gathered user pain points through surveys, interviews, and observations with finance teams and executives.

- Define: Identified core problems: data fragmentation, slow report generation, and lack of visual KPI summaries.
- Ideate: Brainstormed solutions focusing on automated reporting, predictive forecasting, and interactive dashboards.
- Prototype: Designed user interface mockups (UI/UX) for executive dashboards and analyst workflows.
- Test: Collected feedback from potential users to refine system usability and functionality.

2.3 Product Perspective

The software operates as a centralized web platform that integrates with existing corporate databases and legacy IT systems. It transforms raw financial data into structured analytics and interactive reports without replacing core accounting databases.

2.4 User Characteristics

- 1_ Finance Teams: Medium technical skill; need simple tools for daily data input and routine automated reporting.
- 2_ Financial Analysts: High analytical skill; require advanced tools for data extraction, custom modeling, and forecasting.
- 3_ Managers & Executives: Limited time; need intuitive, high-level visual dashboards with key metrics to make fast strategic decisions.

3. Functional Requirements

- 1_ The system shall allow users to log in securely using Role-Based Access Control (RBAC).
- 2_ The system shall import and aggregate financial data from existing databases and spreadsheet files.
- 3_ The system shall generate automated financial reports (e.g., balance sheets, income statements, and cash flow).
- 4_ The system shall display real-time interactive dashboards with Key Performance Indicators (KPIs) for executives.
- 5_ The system shall provide predictive forecasting tools to analyze future revenue and budget trends for financial analysts.
- 6_ The system shall allow users to export generated reports and analytics into multiple formats (PDF, Excel, CSV).
- 7_ The system shall allow administrators to manage user accounts, assign roles, and configure system permissions.
- 8_ The system shall log all user activities and data changes for security auditing purposes.

4. Non-Functional Requirements

- 1_ Performance: The system shall load dashboard metrics and financial charts within 3 seconds under normal network conditions.
- 2_ Security: The system shall encrypt sensitive financial data both in transit (using HTTPS/TLS) and at rest (using AES-256 encryption).
- 3_ Usability: The interface shall feature an intuitive, clear layout requiring minimal training for finance users and executives.

4_ Scalability: The architecture shall support concurrent usage by multiple department teams without performance degradation.

5_ Availability: The system shall maintain an uptime of 99.5% during core business hours to ensure uninterrupted access to financial reports.

6_ Reliability & Data Integrity: The system shall automatically back up financial data daily to prevent data loss.

5. Constraints and Assumptions

5.1 Constraints

1. **Security & Compliance:** The system must strictly adhere to corporate data privacy laws and financial compliance standards.
2. **System Integration:** The software must be compatible with existing corporate legacy databases and standard spreadsheet file formats (.XLSX, .CSV).
3. **Browser Compatibility:** The web-based interface must render correctly on modern web browsers (e.g., Google Chrome, Microsoft Edge, Mozilla Firefox).
4. **Budget & Schedule:** Project development and UI/UX prototyping must be completed within the allocated academic/semester timeline.

5.2 Assumptions

1. **Network Connectivity:** Users have a stable internet or internal corporate network (Intranet) connection to access the platform.
2. **User Capability:** End-users (finance personnel and managers) possess basic computer literacy and familiarity with financial concepts.
3. **Data Availability:** Corporate IT teams will provide clean, structured sample data for testing and system evaluation.
4. **Role Permissions:** Organization administrators will define and manage exact user roles and access rights prior to deployment.

6. Validation and Testing

Usability Testing: Conduct hands-on testing sessions with representative users (finance staff, analysts, and managers) to evaluate dashboard clarity and ease of navigation.

Functional Testing: Verify that all core functional features (data import, report generation, RBAC, and forecasting) operate accurately as specified.

Security & Access Control Testing: Validate Role-Based Access Control (RBAC) to ensure unauthorized users cannot access sensitive financial metrics or export restricted reports.

Performance & Stress Testing: Measure system response times and page load speeds under multi-user access to ensure dashboards render within the 3-second threshold.

Data Accuracy Verification: Cross-check generated financial summaries and exported reports against original source datasets to ensure zero calculation discrepancy.

[Test Cases file Link](#)

7. Conclusion

7.1 Summary

This document outlines the core requirements, scope, and system design for the Financial Analytics and Reporting System. Developed using a Design Thinking approach, the platform addresses key operational challenges such as data fragmentation, manual errors, and delayed executive reporting. By combining automated reporting, predictive forecasting, and interactive real-time dashboards with robust Role-Based Access Control, the software delivers a central, secure solution tailored to finance teams, analysts, and decision-makers.

7.2 Next Steps

Design Thinking – Ideation & Prototyping: Finalize low/high-fidelity UI/UX wireframes for executive dashboards and analyst modules based on initial Empathize phase insights.

Architecture & Database Setup: Establish the core database schema, API integrations for spreadsheet parsing, and security/RBAC protocols.

Core Feature Development: Implement data ingestion pipelines, automated report generation engines, and real-time chart visualization components.

Testing & Validation: Execute test cases (usability, functional, and security) with key stakeholders to refine functionality prior to final deployment.

8. References

Project Source Code & Repository: Fin-Cash Financial Analytics & Management System Codebase.

Repository Link: [Fin-Cash Official GitHub Project](#)

Design Thinking Methodology Framework: Stanford University d.school Design Thinking Guidelines (Empathize, Define, Ideate, Prototype, Test).

Software Engineering Specification Standards: ISO/IEC/IEEE 29148:2018 - Systems and software engineering Requirements Engineering (SRS Standard).

User Research & Project Artifacts: Internal team documentation, Empathize survey notes, and user interview transcripts for Fin-Cash.

Access via project documentation folder in the [Fin-Cash Repository](#).